# Junho Andrew Choi

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## SUMMARY

Data Science and Integrative Biology senior at UC Berkeley targeting machine learning, applied AI, and AI evaluation roles. Two research internships building and stress-testing production AI systems: large language model evaluation on a live enterprise system, and deep learning for 3D medical imaging. Python, SQL, PyTorch, knowledge graphs, model evaluation.

## EDUCATION

**University of California, Berkeley** Expected May 2027
*B.A. Data Science (Concentration: Computational Methods in Biology) and B.A. Integrative Biology* *Berkeley, CA*
**Coursework:** Machine Learning (CS 189, in progress), Data Engineering (DS C101, in progress), Principles & Techniques of Data Science, Data Structures, Probability for Data Science, Optimization Modeling

## SKILLS

**Languages:** Python, SQL, Java, R
**Machine Learning & AI:** PyTorch, scikit-learn, NumPy, pandas, deep learning, large language model (LLM) evaluation, retrieval-augmented generation (RAG), natural language to SQL (Text2SQL), knowledge graphs (Neo4j), model interpretability (SHAP, Grad-CAM), feature engineering
**Data & Analytics:** PostgreSQL, exploratory data analysis (EDA), statistics, data pipelines, model evaluation, data visualization, Jupyter, Git, React, Vite, Figma
**Spoken:** Korean (native), English (fluent), Mandarin

## EXPERIENCE

**SmartMind AI** May 2026 - Aug 2026
*AI Researcher, Data Science Team* *Seoul, Republic of Korea*
- Initiated and led a proof-of-concept natural-language-to-SQL pipeline on a live enterprise resource planning (ERP) system, redesigning evaluation from single-run pass/fail into a 43-run reproducibility framework that converted an unverified 20/20 pass rate into a measured 13/15 and surfaced 3 systemic accounting-logic defects producing valid but silently wrong SQL
- Improved retrieval quality in a Neo4j-based ontology pipeline (ApeRAG/Qurify) by auditing entity-merge accuracy and tuning similarity confidence thresholds, tracing failed queries back to missing or duplicate graph nodes, and migrating 15 business rules from long-form prompts into structured graph nodes

**Asan Medical Center (AMC)** May 2025 - Aug 2025
*AI Research Intern* *Seoul, Republic of Korea*
- Co-authored a research paper on a multimodal deep survival model for hepatocellular carcinoma, developing and fine-tuning 3D U-Net and ResNet3D architectures in PyTorch and integrating SHAP and Grad-CAM to validate feature importance for high-stakes clinical prediction
- Ran large-scale exploratory analysis across 3D CT and clinical modalities for the VLM3D Multi-Abnormality Classification Challenge, informing the feature selection and strategy that placed 7th among multi-institutional teams

**Invention Corps of Berkeley - Mozilla Firefox** Aug 2025 - Feb 2026
*Design Engineer Consultant* *Berkeley, CA*
- Engineered a Firefox performance-monitoring extension in React and Vite that tracked, filtered, and sorted 50+ concurrent tabs by live memory and CPU usage
- Led survey-based user research with a 12-person team, translating findings into feature requirements and Figma interface designs that shaped the shipped extension

## PROJECTS

- **rad-DINO:** implemented and ran experiments with a self-supervised vision transformer for radiology imaging, evaluating learned representations for downstream classification in PyTorch
- **Multi-Agent Ad Generation System:** built an 8-agent large language model pipeline for on-brand ads, where agents exchange a versioned JSON spec rather than prose and a brand-guardian agent hard-vetoes on deterministic color and layout gates
- **AI Tactical Assistant:** trained a PyTorch network to predict match outcomes from historical player and team data, paired with a PuLP linear-programming optimizer generating constrained optimal formations
- **Activation Domain Prediction Pipeline:** developed a CNN-BiLSTM model classifying transcriptional activation domains from protein sequences with sliding-window feature engineering and K-means clustering, reaching F1 = 0.95 across 10 subclasses

## LEADERSHIP & AWARDS

**US Army, 8th Army 2ID HHBN** Sep 2022 - Mar 2024
*Non-Commissioned Officer* *Republic of Korea*
- Led a 40+ soldier platoon as a bilingual Korean-English platoon leader, delivering Army Combat Fitness Test (ACFT) training and coursework to a 96% platoon pass rate
- Awarded the US Army Certificate of Achievement (Mar 2024) for leadership and bilingual communication